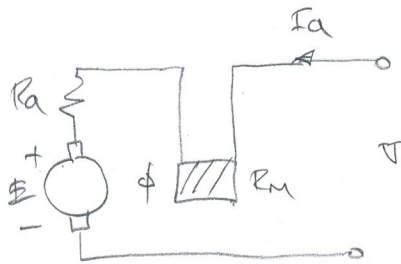


19-8)



$$\begin{aligned} V &= 240 \text{ V} \\ R_a &= 0.42 \Omega \\ R_m &= 0.18 \Omega \\ n_1 &= 500 \text{ RPM} \\ I_{a1} &= 36 \text{ A} \\ I_{a2} &= 21 \text{ A} \\ U_B &= 3 \text{ V} \end{aligned}$$

Bestäm varvtalen då motorn belastas så att strömmen $I_{a2} = 21 \text{ A}$. Φ proportionellt varvtalen.

Lösning: Vi vet att

$$E_1 = k_e \Phi \omega_1 \quad (1)$$

KSPLUR

$$-V + (R_m + R_a)I_{a1} + U_B + E = 0 \quad (2)$$

$$(1), (2) \Rightarrow \frac{V - (R_m + R_a)I_{a1} - U_B}{\omega_1} = k_e \Phi_1$$

NUMERISKT

$$k_e \Phi_1 = \frac{240 - (0.42 + 0.18)36 - 3}{2\pi \cdot \frac{500}{60}} = 4.1138$$

VARVTAL 2: AN

$$\omega_2 = \frac{V - (R_m + R_a)I_{a2} - U_B}{k_e \Phi_2} =$$

$$= \frac{240 - 0.6 \cdot 21 - 3}{\frac{21}{36} \cdot 4.1138} = 93.51 \text{ RAD/sek}$$

$$\approx 892.96 \text{ RPM}$$

SVAR: $n_2 = 892.9 \frac{\text{VARV}}{\text{MIN}}$

ORR!

$$\Phi_1 = k I_{a1}$$

$$\Phi_2 = k I_{a2}$$

$$\Phi_2 = \frac{I_{a2}}{I_{a1}} \Phi_1$$